

# Day 3 Lecture Notes: Iteration and Eigenvalues

## Mathematical Modeling & Computational Projects Camp

### Morning Structure (10:00–12:30)

- 10:00–10:30: Lecture Block 1
- 10:35–11:05: Problem Solving Session 1 (30 minutes)
- 11:10–11:40: Lecture Block 2
- 11:40–12:10: Problem Solving Session 2 (shorter) + 12:10–12:30 synthesis/review

### Lecture Block 1 (10:00–10:30): Iterative Models

If

$$\mathbf{x}_{k+1} = A\mathbf{x}_k,$$

then each step applies the same update rule. Iteration is useful for repeated processes over time.

Example context: student preferences switching between two clubs each semester.

Core vocabulary:

- **State vector**  $\mathbf{x}_k$ : system snapshot at time  $k$ .
- **Transition matrix**  $A$ : rule that updates one step.
- **Initial condition**  $\mathbf{x}_0$ : where we start.

Instructor prompt: “Why do we need both  $A$  and  $\mathbf{x}_0$ ?”

Worked example 1:

$$A = \begin{bmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{bmatrix}, \quad \mathbf{x}_0 = \begin{bmatrix} 100 \\ 50 \end{bmatrix}.$$

Compute one step:

$$\mathbf{x}_1 = A\mathbf{x}_0 = \begin{bmatrix} 0.7(100) + 0.2(50) \\ 0.3(100) + 0.8(50) \end{bmatrix} = \begin{bmatrix} 80 \\ 70 \end{bmatrix}.$$

Interpretation: after one update, category 1 has 80 and category 2 has 70.

Worked example 2 (second step):

$$\mathbf{x}_2 = A\mathbf{x}_1 = \begin{bmatrix} 0.7(80) + 0.2(70) \\ 0.3(80) + 0.8(70) \end{bmatrix} = \begin{bmatrix} 70 \\ 80 \end{bmatrix}.$$

Observation to discuss: some iterative models stabilize, oscillate, or grow.

## Problem Solving Session 1 (10:35–11:05): Practice Set A (30 minutes)

**Goal:** Compute iterative updates accurately and interpret matrix entries.

Suggested pacing:

- First 10 minutes: Problems 1–4.
- Next 12 minutes: Problems 5–8.
- Last 8 minutes: Problems 9–10 and reasoning checks.

1. For  $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$  and  $\mathbf{x}_0 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ , compute  $\mathbf{x}_1$ .
2. Continue Problem 1 to compute  $\mathbf{x}_2$ .
3. For  $A = \begin{bmatrix} 0.8 & 0.1 \\ 0.2 & 0.9 \end{bmatrix}$  and  $\mathbf{x}_0 = \begin{bmatrix} 100 \\ 50 \end{bmatrix}$ , compute one step.
4. Explain one sentence interpretation of entry 0.2 in row 2, column 1.
5. Why does iteration need an initial state?
6. Give one non-math real-world process that is iterative.
7. If two consecutive states become almost equal, what might that suggest?
8. In a two-state model, why should states stay nonnegative?
9. Let  $A = \begin{bmatrix} 0.6 & 0.4 \\ 0.1 & 0.9 \end{bmatrix}$  and  $\mathbf{x}_0 = \begin{bmatrix} 120 \\ 80 \end{bmatrix}$ . Compute  $\mathbf{x}_1$  and  $\mathbf{x}_2$ .
10. Reflection: give one model assumption that could make long-term predictions unreliable.

Instructor checkpoint: ask one team to interpret each entry of  $A$  in words before any multiplication.

## Lecture Block 2 (11:10–11:40): Eigenvalue Intuition

A nonzero vector  $\mathbf{v}$  is an eigenvector of  $A$  if

$$A\mathbf{v} = \lambda\mathbf{v}.$$

Interpretation: direction stays the same, size scales by  $\lambda$ .

If  $|\lambda| < 1$ , that direction shrinks over repeated updates. If  $|\lambda| > 1$ , that direction grows.

Geometric intuition for students:

- Most vectors change direction under  $A$ .
- Eigenvectors are special directions that do not turn.
- Eigenvalue tells how fast stretching/shrinking happens.

Worked check example:

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}, \quad \mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Then

$$A\mathbf{v}_1 = 2\mathbf{v}_1, \quad A\mathbf{v}_2 = 3\mathbf{v}_2.$$

So  $\mathbf{v}_1, \mathbf{v}_2$  are eigenvectors with eigenvalues 2 and 3.

Connection to long-term behavior: in many models, the dominant eigenvalue drives which pattern eventually dominates.

## Problem Solving Session 2 (11:40–12:10): Practice Set B (shorter)

**Goal:** Identify eigenvectors/eigenvalues and connect them to iteration behavior.

Suggested pacing:

- First 12 minutes: Problems 1–4.
- Next 10 minutes: Problems 5–7.
- Last 8 minutes: Problems 8–9 and class summary.

1. Check whether  $\mathbf{v} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$  is an eigenvector of  $\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$ .

2. Check whether  $\mathbf{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$  is an eigenvector of  $\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$ .

3. In your own words, define eigenvalue.

4. Why are eigenvalues useful in long-term prediction?

5. Give one assumption in a two-state transition model.

6. For  $A = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}$ , find eigenvalue for  $\mathbf{v} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ .

7. For the same  $A$ , what happens to the first component after many iterations if that direction is present?

8. Suppose an eigenvalue is 0.6. What does that suggest about repeated updates in that direction?

9. Reflection: explain why checking  $A\mathbf{v}$  against a multiple of  $\mathbf{v}$  is the key test.

## Synthesis and Review (12:10–12:30)

- Compare one iterative calculation and one eigenvector check.
- Link to applications: Markov chains and ranking systems.
- Exit check: “What does  $A\mathbf{x}$  mean in words?”
- Exit check 2: “What does  $|\lambda| < 1$  tell us about long-term behavior?”
- Exit check 3: “Why can a model fit one step well but still fail in the long run?”

## Python Preview (Afternoon)

Simulate repeated updates and compute eigenpairs with `numpy.linalg.eig`.